**AGA KHAN UNIVERSITY EXAMINATION BOARD**

**SECONDARY SCHOOL CERTIFICATE**

**CLASS X**

**ANNUAL EXAMINATIONS 2021**

**Physics**

**Total Time: 1 hour 40 minutes**

**Total Marks: 50 (40-Theory & 10-Alternate to Practical)**

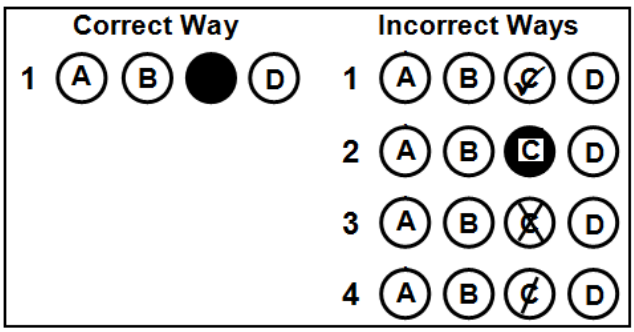
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## INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 50 only.
4. Question Distribution:

| Theory | Alternate to Practical (ATP) |
|---|---|
| 40 MCQs | 10 MCQs |

5. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.




**Candidate's Signature**

6. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
7. DO NOT write anything in the answer grid. The computer only records what is in the circles.
8. The marks obtained on the 40 MCQs will be equated to the total marks of 65 for the theory examination results.
9. You may use a simple calculator if you wish.

## **THEORY** (Questions 1-40)

1. If the particles of a medium vibrate in the direction of propagation of wave, then it is called a/ an

   A. surface wave.
   B. transverse wave.
   C. longitudinal wave.
   D. electromagnetic wave.

2. If the frequency of a wave is doubled, then the time period of the wave will

   A. become double.
   B. decrease to half.
   C. remain the same.
   D. decrease to a quarter.

3. If the time period of a simple pendulum is 2 s on the surface of the Earth, then the time period of the same pendulum on the surface of the Moon will be

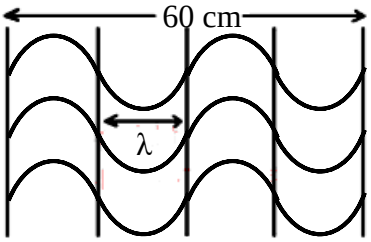
   A. greater than 2 s.
   B. equal to 2 s.
   C. less than 2 s.
   D. equal to 0 s.

4. If a mechanical wave moves from a rare to the denser medium, then which of the following factors will change during refraction?

   I. Wavelength
   II. Speed of wave
   III. Frequency of wave

   A. I only
   B. III only
   C. I and II
   D. II and III

5. In the given figure, the frequency of the wave source is 3 Hz.




Waves

The velocity of the wave will be

   A. 0.5 cm/s.
   B. 12 cm/s.
   C. 20 cm/s.
   D. 45 cm/s.

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6. Sound energy passing per second through a unit area held perpendicular to the direction of propagation of sound waves is called the

   A. pitch of sound.
   B. quality of sound.
   C. intensity of sound.
   D. loudness of sound.

7. When sound is incident on the surface of a medium, it bounces back into the first medium. This phenomenon is called

   A. an echo.
   B. the acoustic.
   C. the loudness of sound.
   D. the amplitude of sound.

8. Which of the following characteristics of a musical sound depends upon the frequency?

   A. Pitch
   B. Quality
   C. Loudness
   D. Amplitude

9. Which of the following optical media is used for the correction of short-sightedness?

   A. Convex lens
   B. Concave lens
   C. Convex mirror
   D. Concave mirror

10. All of the following are the uses of a compound microscope EXCEPT it

    A. is used for observing microorganisms.
    B. helps human beings in the field of medical testing.
    C. helps to find the heavenly bodies in the outer space.
    D. is used in the forensics to examine evidences from a crime scene.

11. An object is placed between the principal focus (F) and the centre of curvature (2F).

    An image will be formed on the other side of the convex lens

    A. at F.
    B. at 2F.
    C. beyond 2F.
    D. between F and 2F.

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12. Which of the following is/ are the condition(s) necessary for the total internal reflection?

I. Angle of incidence is always greater than the critical angle.
II. Angle of incidence is always equal to the angle of refraction.
III. Light rays should always travel from a rare medium to the denser medium.

A. I only
B. III only
C. I and II
D. II and III

13. An image is formed at a distance of 32 cm from a concave mirror which has a focal length of 12 cm. The object distance is

A. – 0.052 cm.
B. – 19.2 cm.
C. 20.0 cm.
D. 32.0 cm.

14. The floor of a water tank appears at a depth of 3 m. If the refractive index of water is 1.33, then the actual depth of water is

A. 0.44 m.
B. 1.67 m.
C. 2.25 m.
D. 3.99 m.

15. Compared to the eye piece of a compound microscope, the focal length of the objective lens of an astronomical telescope is

A. larger.
B. smaller.
C. infinite.
D. negative.

16. Which of the following statements is TRUE about the image formed by a convex mirror?

A. It is real.
B. It is inverted.
C. It is smaller than the object.
D. It is always formed at the principal focus.

17. In a compound microscope, the focal length of the objective lens as compared to the eye piece is always

A. larger.
B. smaller.
C. infinite.
D. of same size.

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18. Two capacitors of capacitance 6 µF and 3 µF are connected in series across a cell of 18 V.

The equivalent capacitance will be

A. 2 µF.
B. 3 µF.
C. 6 µF.
D. 9 µF.

19. All of the following are the precautions which should be taken during installation of any electrical equipment EXCEPT

A. turn off any attached peripherals.
B. attach the electrical equipment to AC power.
C. wear shoes with non-conductive rubber soles.
D. disconnect any telecommunication lines from the system.

20. Photocopier and inkjet printers are the applications of

A. electricity.
B. electronics.
C. electrostatics.
D. electromagnetism.

21. The direction of electric intensity to an electric field can be indicated by drawing lines. These lines are known as

A. electric field lines.
B. magnetic field lines.
C. magnetic field intensity.
D. electromagnetic field intensity.

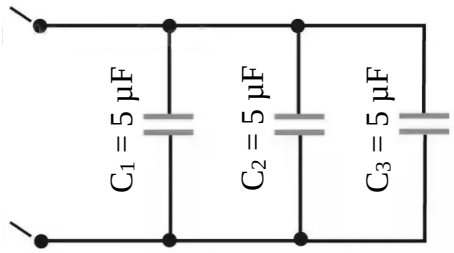
22. The electric force of repulsion between two electrons at a distance of 1 m is

(**Note**: The value of the constant of proportionality is $9 \times 10^9$ $Nm^2/C^2$ and the charge on electron is $1.6 \times 10^{-19}$ C.)

A. $1.29 \times 10^{-1}$ N.
B. $1.44 \times 10^{-9}$ N.
C. $2.30 \times 10^{-28}$ N.
D. $2.30 \times 10^{-30}$ N.

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23. The equivalent capacitance of the given combination of the capacitors will be




A. 0.067 µF.
B. 0.6 µF.
C. 5 µF.
D. 15 µF.

24. All electrical appliances are connected parallel to each other between the live and neutral wires.

In light of the given statement, which of the following options is TRUE for electric current and potential difference?

| | **Electric Current** | **Potential Difference** |
|---|---|---|
| A | Remain the same | Remain the same |
| B | Remain the same | Varies |
| C | Varies | Remain the same |
| D | Varies | Varies |

25. If three resistors of 5 Ω each are connected to form a triangle, then the resistance between any two vertices (terminals) will be

A. 0.3 Ω.
B. 3.333 Ω.
C. 5 Ω.
D. 15 Ω.

26. If the length and diametre of a solid cylindrical shape conductor is doubled, then the resistance will

A. reduce to half.
B. reduce to quarter.
C. increase two times.
D. increase four times.

27. If three equal resistors and a battery are given, then how many different combinations of workable circuits can be made?

A. Three
B. Four
C. Five
D. Six

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28. Consider two metallic conductors **A** and **B**. Suppose, the length of a metallic conductor **A** is increased and the cross-sectional area of the metallic conductor **B** is also increased.

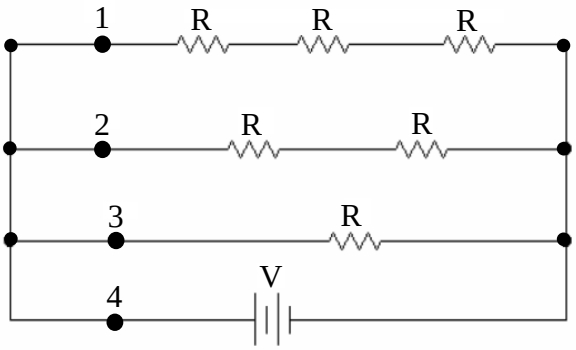
What will be the effect on the resistance of both conductors?

| | **Resistance of the** | |
|---|---|---|
| | **Conductor A** | **Conductor B** |
| A | Increases | Increases |
| B | Increases | Decreases |
| C | Decreases | Increases |
| D | Decreases | Decreases |

29. The energy dissipated by a 100 W bulb in 60 s is

A. 1600 J.
B. 3200 J.
C. 3600 J.
D. 6000 J.

30. Which of the following points (1 to 4) in the given circuit will receive the lowest magnitude of current?




A. 1
B. 2
C. 3
D. 4

31. The purpose of using a transformer in an electric circuit is to

A. alter the electrical power.
B. adjust the electrical energy.
C. fluctuate the magnetic field.
D. change the voltage or current.

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32. The direction of force on a current carrying conductor placed in a magnetic field can be determined by

A. Joule's law.
B. Coulomb's law.
C. Head to tail rule.
D. Fleming's left-hand rule.

33. If a conductor passes through a magnetic field twice with different speeds, then which of the following option is CORRECT with respect to speed and induced electromotive force (e.m.f.)?

| | **Speed** | **Induced Electromotive Force** |
|---|---|---|
| A | High | Increased |
| B | Low | Unchanged |
| C | Low | Increased |
| D | High | Decreased |

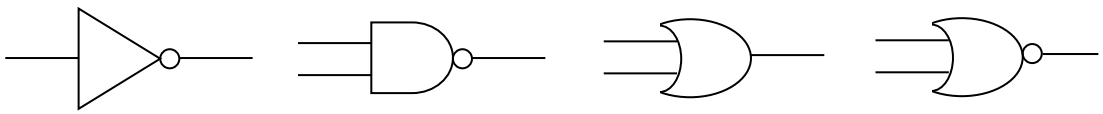
34. If the current through a coil of an electromagnet reverses, then the

A. magnetic field increases.
B. magnetic field decreases.
C. direction of the magnetic field reverses.
D. direction of the magnetic field remains the same.

35. Electromagnets are commonly used in all of the following electrical instruments EXCEPT

A. iron.
B. toaster.
C. motors.
D. generator.

36. Which of the following is the symbol of a NAND gate?



A B C D

37. Cathode Rays Oscilloscope (CRO) is NOT used

A. as a voltmeter.
B. to measure alternating and direct current.
C. to measure the effect of background radiation.
D. to study the effect of electric field on moving charges.

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38. All of the following are the fundamental gates EXCEPT

A. OR.
B. NOT.
C. NOR.
D. AND.

39. In thermionic emission, when a filament is heated, a beam of tiny particles is emitted from it. These particles are known as

A. protons.
B. electrons.
C. beta particles.
D. alpha particles.

40. In electronics, the characteristics of an analogue quantity is that it

A. randomly selects sets of quantities.
B. can be represented by discrete levels.
C. has a continuous set of values over a given range.
D. can be described with a finite number of mathematical steps.

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## ALTERNATE TO PRACTICAL (ATP: Questions 41-50)

41. In a science lab, Rubia and Kiran performed the experiment of simple pendulum individually.

They used the same metallic bob, but they got different results of time period.

This is because they

A. have taken different amplitudes.
B. used different thickness of threads.
C. have taken different lengths of the pendulum.
D. performed the experiment at different positions.

42. A simple pendulum of 1 m length is placed at a location where acceleration due to gravity is 10 $m/s^2$. Its time period will be

A. 0.50 s.
B. 0.99 s.
C. 1.98 s.
D. 3.96 s.

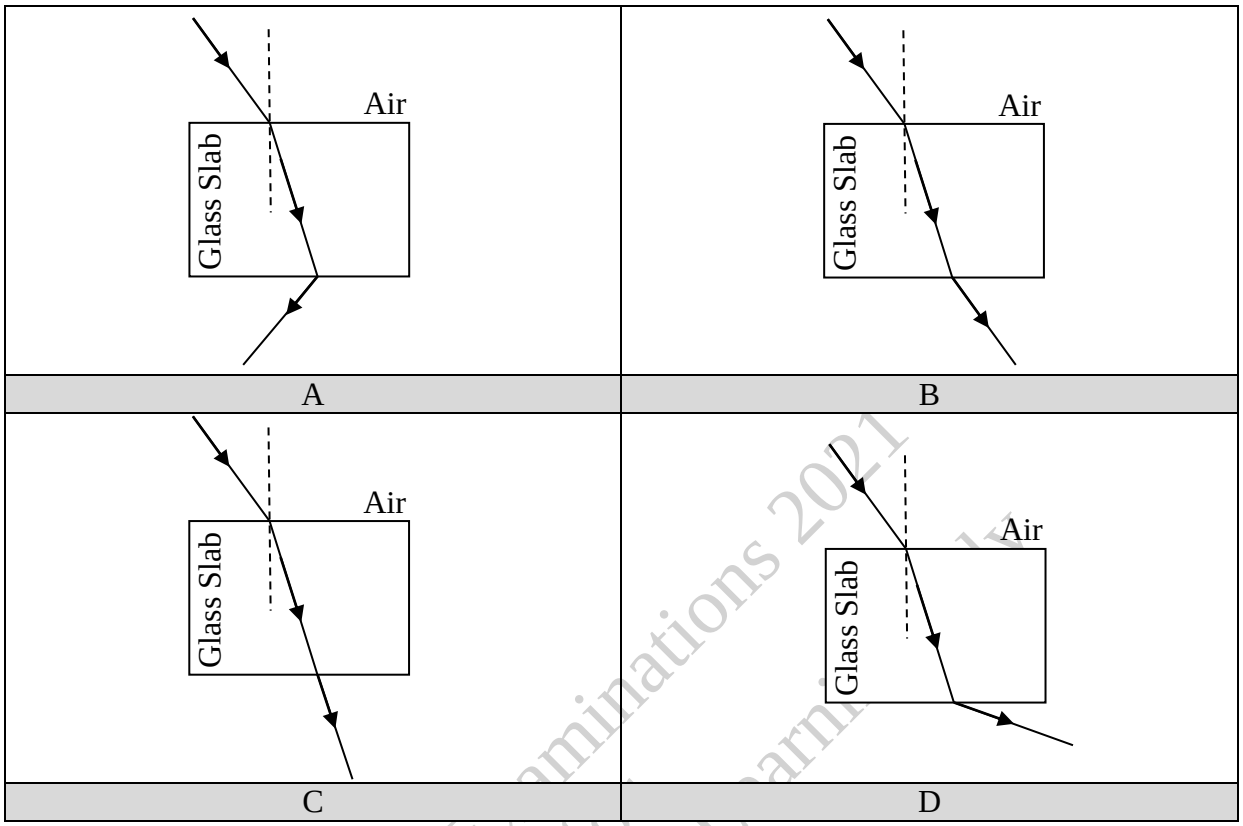
43. Qasim, a student of grade X has been given a task by his teacher to find the image of an object that should be real, inverted and of the same size by using a concave mirror.

To fulfil the task, Qasim should place the object

A. at the principal focus.
B. at the centre of curvature.
C. beyond the centre of curvature.
D. between the principal focus and the centre of curvature.

PLEASE TURN OVER THE PAGE

44. Which of the following is the representation of the refraction of light?




45. An object is placed in front of a convex lens which has a focal length of 15 cm. A virtual image is formed at a distance of 10 cm.

The object distance will be

A. – 30 cm.
B. – 25 cm.
C. 5 cm.
D. 6 cm.

46. When a very high resistance is connected in series with a galvanometer than it will convert into

A. ammeter.
B. voltmeter.
C. ohmmeter.
D. potentiometer.

47. If in a series circuit numbers of bulbs are increased, then it will

A. increase the voltage in the circuit.
B. decrease the brightness of the bulbs.
C. decrease the resistance in the circuit.
D. increase the flow of the electric current.

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48. All of the following statements are TRUE about the magnetic field lines EXCEPT that they

A. do not intersect each other.
B. form a continuous closed curve.
C. appear in the direction of tangent at any point on the magnetic field lines.
D. travel from South to North Pole outside the current carrying circular coil.

49. If an AND gate is connected in a circuit with two inputs **A** and **B** to light up a bulb, the bulb will glow, when the condition of input **A** and **B** is

| | **Input A** | **Input B** |
|---|---|---|
| A | off | on |
| B | off | off |
| C | on | on |
| D | on | off |

50. The given table depicts the truth table of an OR gate with some errors.

| | **Inputs** | | **Output** |
|---|---|---|---|
| | **P** | **Q** | **R** |
| I | 0 | 1 | 0 |
| II | 1 | 0 | 0 |
| III | 0 | 0 | 0 |
| IV | 1 | 1 | 1 |

Which of these rows is/ are error free?

A. IV only
B. I and II
C. III and IV
D. I, II and III

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